

Week 3 Assignment Solutions

Question 1: Brownian Motion Paths and Quadratic Variation

(a) The Four Defining Properties of Brownian Motion

1. **Initial Value** ($W_0 = 0$): *Financial Interpretation:* We start measuring our cumulative asset returns or shocks from a baseline of zero today.
2. **Independent Increments** ($W_t - W_s$ is independent of $\mathcal{F}_s$): *Financial Interpretation:* The market is efficient; knowing the entire past price history gives you absolutely no predictive advantage for future price movements.
3. **Stationary Gaussian Increments** ($W_t - W_s \sim \mathcal{N}(0, t - s)$): *Financial Interpretation:* Price shocks are normally distributed, have an expected return of zero, and their variance scales perfectly linearly with time (meaning risk scales by the square root of time).
4. **Continuous Paths:** *Financial Interpretation:* Prices do not magically teleport from one value to another; they must continuously trade through all intermediate values.

(b) Derivation of the Covariance Formula

Assume $0 \leq s \leq t$. We want to find $\text{Cov}(W_s, W_t)$.

First, we rewrite W_t by breaking it into two pieces: the path up to s , and the increment from s to t :

$$W_t = W_s + (W_t - W_s)$$

Now, plug this into the covariance operator:

$$\text{Cov}(W_s, W_t) = \text{Cov}(W_s, W_s + (W_t - W_s))$$

Using the linearity of covariance:

$$\text{Cov}(W_s, W_t) = \text{Cov}(W_s, W_s) + \text{Cov}(W_s, W_t - W_s)$$

The term $\text{Cov}(W_s, W_s)$ is simply the variance of W_s , which we know is s .

Because of the **Independent Increments** property, the future movement ($W_t - W_s$) is completely independent of the past movement W_s . Therefore, their covariance is strictly 0.

$$\text{Cov}(W_s, W_t) = s + 0 = s$$

Since we assumed $s \leq t$, we can generalize this for any two times as $\text{Cov}(W_s, W_t) = \min(s, t)$.

(c) Expected Value of Quadratic Variation

Let $Q^n = \sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2$. Using the linearity of expectation:

$$E[Q^n] = \sum_{k=0}^{n-1} E[(W_{t_{k+1}} - W_{t_k})^2]$$

We know that the increment $(W_{t_{k+1}} - W_{t_k}) \sim \mathcal{N}(0, \Delta t_k)$. The expected value of a squared zero-mean variable is simply its variance. Therefore, $E[(W_{t_{k+1}} - W_{t_k})^2] = \Delta t_k$.

$$E[Q^n] = \sum_{k=0}^{n-1} \Delta t_k = T$$

(d) Variance of Quadratic Variation and Convergence

Because the increments of Brownian motion are independent, the *squared* increments are also independent. Therefore, the variance of the sum is the sum of the variances:

$$\text{Var}(Q^n) = \sum_{k=0}^{n-1} \text{Var}((W_{t_{k+1}} - W_{t_k})^2)$$

For a normal variable $Z \sim \mathcal{N}(0, \sigma^2)$, the variance of its square is $2\sigma^4$. Here, $\sigma^2 = \Delta t_k$, so the variance is $2(\Delta t_k)^2$.

$$\text{Var}(Q^n) = \sum_{k=0}^{n-1} 2(\Delta t_k)^2$$

We can pull out the maximum step size $\|\Pi\|$:

$$\text{Var}(Q^n) \leq 2\|\Pi\| \sum_{k=0}^{n-1} \Delta t_k = 2\|\Pi\|T$$

As $\|\Pi\| \rightarrow 0$, the upper bound goes to 0, forcing $\text{Var}(Q^n) \rightarrow 0$.

Since the expected value is exactly T and the variance shrinks to 0, Q^n converges to T in L^2 .

Remarkability: It is mathematically astonishing that a completely random, chaotic, and unpredictable path perfectly accumulates its squared movements into a strictly deterministic, fixed number (T).

(e), (f), (g) Computational Observations

- **Part (e) Observation:** The empirical quadratic variation will be very close to 1.0 (e.g., 0.998). In stark contrast, the total variation will be massive (e.g., 80+). This confirms that a Brownian path is infinitely jagged (infinite total distance traveled) but tightly constrained in its squared variance.
- **Part (f) Observation:** The plot of empirical QV vs. N shows high volatility at low N (e.g., $N = 10$), but tightly converges to the horizontal $T = 1$ line as N grows. The rate of convergence is slow because the standard error scales by $\mathcal{O}(\sqrt{\Delta t})$.
- **Part (g) Observation:** For $t = 0.5$, the empirical mean of the 10,000 paths converges to approximately 0, and the empirical variance converges to exactly 0.5, perfectly matching the theoretical $\mathcal{N}(0, t)$ marginal distribution.

Question 2: Itô's Lemma and Geometric Brownian Motion

(a) Itô's Lemma

For an SDE $dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t$, Itô's Lemma states:

$$df(t, X_t) = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(dX_t)^2$$

- $\frac{\partial f}{\partial t}dt$: The deterministic time derivative.
- $\frac{\partial f}{\partial x}dX_t$: The first-order chain rule term.
- $\frac{1}{2}\frac{\partial^2 f}{\partial x^2}(dX_t)^2$: The Itô correction term.

Why modified? In ordinary calculus, $(dx)^2 \rightarrow 0$ infinitely fast. But for a stochastic process, $(dW_t)^2 = dt$. Because squared random increments scale with time, the second-order Taylor expansion term does not vanish and must be included.

(b) Deriving the Explicit Solution for GBM

Let $f(x) = \ln(x)$. We find the derivatives: $\frac{\partial f}{\partial x} = \frac{1}{x}$ and $\frac{\partial^2 f}{\partial x^2} = -\frac{1}{x^2}$. Applying Itô's Lemma to $f(S_t) = \ln(S_t)$:

$$d(\ln S_t) = \frac{1}{S_t}dS_t + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)(dS_t)^2$$

Substitute $dS_t = \mu S_t dt + \sigma S_t dW_t$ and note that $(dS_t)^2 = \sigma^2 S_t^2 dt$:

$$d(\ln S_t) = \frac{1}{S_t}(\mu S_t dt + \sigma S_t dW_t) - \frac{1}{2S_t^2}(\sigma^2 S_t^2 dt)$$

$$d(\ln S_t) = \mu dt + \sigma dW_t - \frac{\sigma^2}{2}dt = \left(\mu - \frac{\sigma^2}{2}\right)dt + \sigma dW_t$$

Integrating both sides from 0 to t :

$$\ln(S_t) - \ln(S_0) = \left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t$$

Taking the exponential of both sides yields the explicit solution:

$$S_t = S_0 \exp \left\{ \left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t \right\}$$

(c) The Itô Correction and Pricing Error

The Itô correction $-\sigma^2/2$ acts as a "volatility drag." It adjusts for the mathematical reality that the expected value of an exponential is greater than the exponential of the expected value (Jensen's Inequality).

Computing the expected value:

$$E[S_t] = S_0 \exp \left\{ \left(\mu - \frac{\sigma^2}{2}\right)t \right\} E[\exp\{\sigma W_t\}]$$

Using the moment generating function for a Normal variable $W_t \sim \mathcal{N}(0, t)$, $E[\exp\{\sigma W_t\}] = \exp\{\sigma^2 t/2\}$:

$$E[S_t] = S_0 \exp \left\{ \mu t - \frac{\sigma^2 t}{2} + \frac{\sigma^2 t}{2} \right\} = S_0 e^{\mu t}$$

If omitted, the incorrect expectation is $E[S_{wrong}] = S_0 e^{\mu t} \cdot e^{\sigma^2 t/2}$.

For $\sigma = 0.2$ and $t = 10$: The error factor is $e^{(0.2^2 \times 10)/2} = e^{0.2} \approx 1.2214$. Omitting the correction results in a massive **22.14%** overestimation of the expected stock price.

(d) - (g) Computational Observations

- **Part (e) Observation:** The empirical mean and standard deviation of S_T perfectly align with the theoretical values of the log-normal distribution, validating the Euler-Maruyama discretization.
- **Part (f) Observation:** As volatility (σ) increases from 0.1 to 0.5, the distribution of S_T becomes wildly right-skewed. The standard deviation explodes. Consequently, the expected option payoff $E[(S_T - K)^+]$ increases dramatically, confirming the financial principle that options are more valuable in highly volatile markets due to their asymmetric payoff profile.
- **Part (g) Observation:** The empirical log-returns perfectly match a Gaussian distribution with mean $(\mu - \sigma^2/2)T$ and variance $\sigma^2 T$, confirming that returns (not absolute prices) are log-normally distributed.

Question 3: Euler-Maruyama Simulation and the Ornstein-Uhlenbeck Process

(a) Drift, Diffusion, and Mean Reversion

- **Drift Coefficient:** $-\theta X_t$
- **Diffusion Coefficient:** σ
- **Financial Meaning:** The drift acts as an algorithmic rubber band pulling the interest rate toward 0. If X_t is highly positive, $-\theta X_t$ becomes a large negative number, dragging the rate down. If X_t is highly negative, $-\theta X_t$ becomes positive, pushing the rate up. It prevents the explosive exponential growth seen in standard stocks.

(b) Euler-Maruyama Discretization and Stability

The discrete step equation is:

$$X_{t_{k+1}} = X_{t_k} - \theta X_{t_k} \Delta t + \sigma \Delta W_k$$

Stability Condition: To prevent the discrete numerical simulation from overshooting the mean and spiraling into infinity, the time step must satisfy $\Delta t < 2/\theta$.

(c) The Stationary Distribution

As $t \rightarrow \infty$, the initial starting position X_0 is entirely "forgotten" by the process, and the distribution of X_t settles into a steady-state Gaussian bell curve $\mathcal{N}(0, \sigma^2/2\theta)$.

Why useful? Unlike stock prices which can theoretically grow to infinity, macroeconomic variables like central bank interest rates fluctuate within specific natural bounds. A stationary distribution perfectly models a rate that jitters randomly but is fundamentally anchored to a long-run economic average.

(g) Strong vs. Weak Convergence

- **Strong Convergence:** Measures how accurately the simulated numerical path tracks the exact true path of a specific reality. It is required for pricing path-dependent derivatives (like Barrier or Asian options).
- **Weak Convergence:** Measures how accurately the final distribution (the histogram) of the simulated paths matches the true theoretical distribution, regardless of how the paths got there.
- **Why weak is sufficient:** Standard European option pricing is just an expected value calculation $E[g(X_T)]$. Because the expectation is just the average of the final terminal distribution, as long as the simulated histogram is accurate (weak convergence), the option price will be mathematically correct.

(d), (e), (f) Computational Observations

- **Part (d) Observation:** The simulated paths all start at 3.0 but are aggressively pulled down to the 0 baseline. Once at the baseline, they jitter randomly but remain tightly confined within the ± 2 standard deviation theoretical bounds.
- **Part (e) Observation:** The histogram of the 20,000 paths at $T = 10$ perfectly matches the theoretical stationary normal density $\mathcal{N}\left(0, \frac{\sigma^2}{2\theta}\right)$, proving the system reached equilibrium.
- **Part (f) Observation:** Increasing θ acts as a stronger rubber band. For $\theta = 5.0$, the process crashes back to zero almost instantly and fluctuates in a very narrow band. For $\theta = 0.5$, the reversion is slow, taking a long time to reach the baseline, resulting in a much wider and flatter stationary distribution.